CS-302: Artificial Intelligence

LECTURE NO 13:

Engr. Tehseen Ahsan
Assistant Professor, Computer Engineering Department
HITEC University Taxila Cantt, Pakistan

Organization of Lecture

1. Learning Process and Components of Learner System.
2. Neuron Biological Model.
3. Artificial Neural Network (ANN).
4. Types of Artificial Neural Network.
5. Activation Functions.

1. Learning Process and Components of Learner System (1/3)

Learning Process:

↳ Learning is defined as specialized form of knowledge acquisition. [Fact 1 → Fact2 → conclusion]

↳ Learning is constructing or modifying representation of what is being experienced.

Components of Learner System:

① **Learning component:** Main component of any learner system and its role is to acquire knowledge and make changes/Improvements to system depending on performance.

② **Performance Component:** Task is performed by choosing actions that needs to be taken.

1. Learning Process and Components of Learner System (2/3)

(3) **Problem Generator:** Suggests Problems/actions that would lead to generation of new examples to improve learning.

(4) **Critic:** Gives feedback to the learner system

Learning can be achieved by:

i) By takeing advice.

ii) From observations.

iii) From examples.

iv) By direct instructions (from an expert).

v) By analyzing differences (i.e., the diffrence b/w the actual outcome versus expected outcome).

1. Learning Process and Components of Learner System (3/3)

vi) By deduction

vii) By explaining experiences.

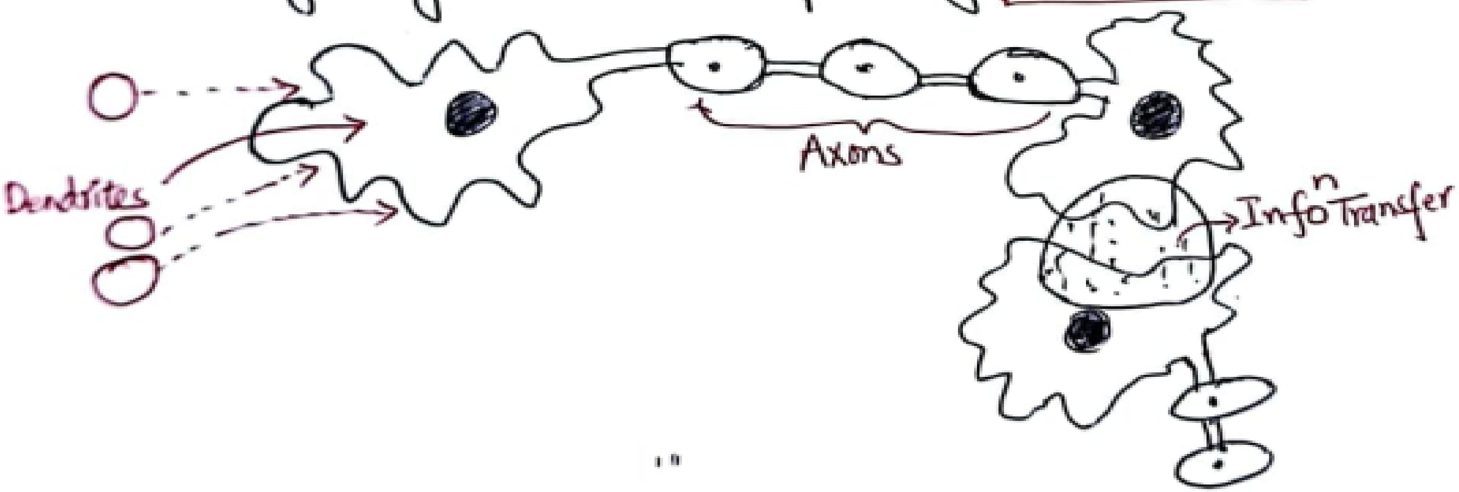
viii) By using relevant Information.

ix) By correcting mistakes.

x) By recording cases (i.e., domain related cases).

2. Neuron Biological Model (1/2)

- Biological Neuron model is also called as Spiking Neuron model.
- Mathematical description of the properties of certain cells in nervous system that generates sharp electrical potentials across their cell membrane.
- Human Brain is composed of Nerve cells called Neurons and they are connected to other thousand cells by Axons.
- Stimulus from external environment or i/p from sensory organs are accepted by Dendrites.



2. Neuron Biological Model (2/2)

Characteristics:

1. Flexibility: Self adjusting network to new environment.

2. Robustness/Fault Tolerance: Decay of Nerve cells doesn't affect Performance.

3. Parallel Computation: Many operations are performed Parallely in distributed fashion.

4. Ability to handle variety of data situations.

3. Artificial Neural Network (ANN) [1/3]

↳ ANN is an information processing paradigm that is inspired by the way of biological nervous system.

↳ ANN is configured for specific application
i) Data classification
ii) Pattern Recognition.

↳ Model
i) Interconnections b/w various neurons.
ii) Activation Function
iii) Learning Rules for data classification & Pattern recognition.

3. Artificial Neural Network (ANN) [2/3]

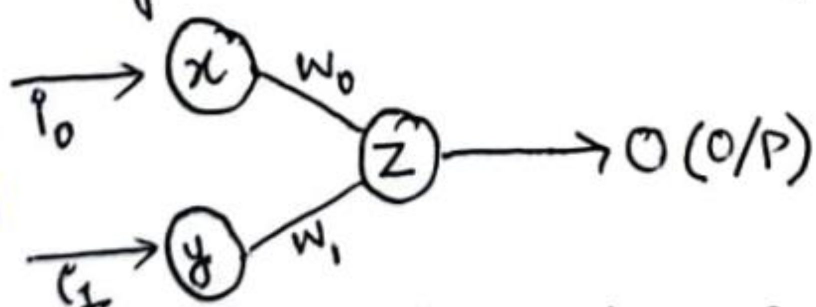
Characteristics of ANN:

i) Neurally implemented mathematical model

ii) Huge no. of interconnected processing elements called neurons for processing.

iii) Input signals arrive at processing elements through connection and connected weights.

Simplest ANN Model

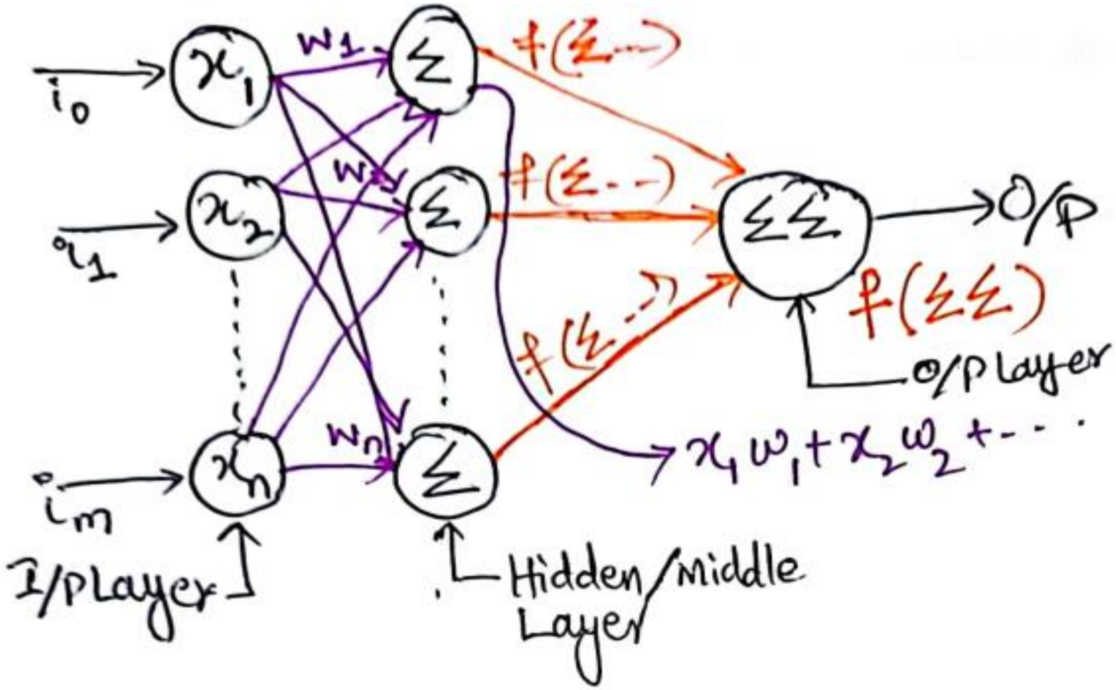


$\text{Input} = i_0 w_0 + i_1 w_1 \Rightarrow$ weighted sum.

$\text{Output} = f(I)$ → Input

→ Activation Function

3. Artificial Neural Network (ANN) [3/3]



Advantages:

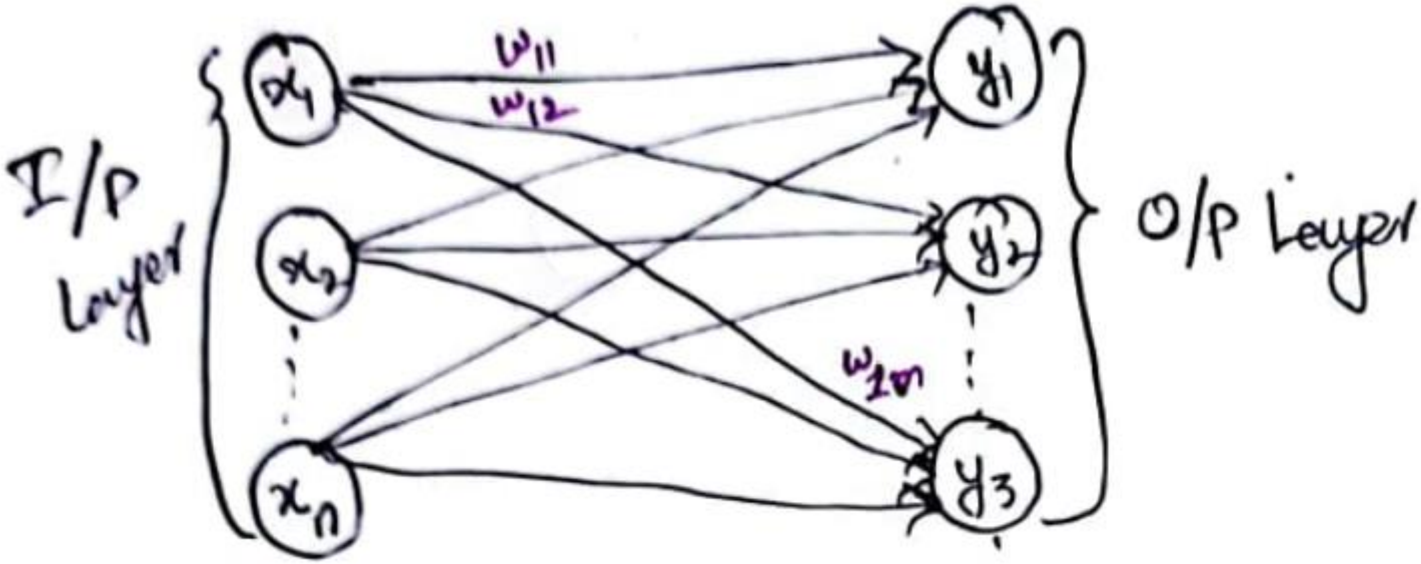
i) It has the ability to learn & model non-linear & complex relationships (with the use of Activation function).

ii) Easy Generalization

iii) No restriction on input variable.

4. Types of Artificial Neural Network (1/4)

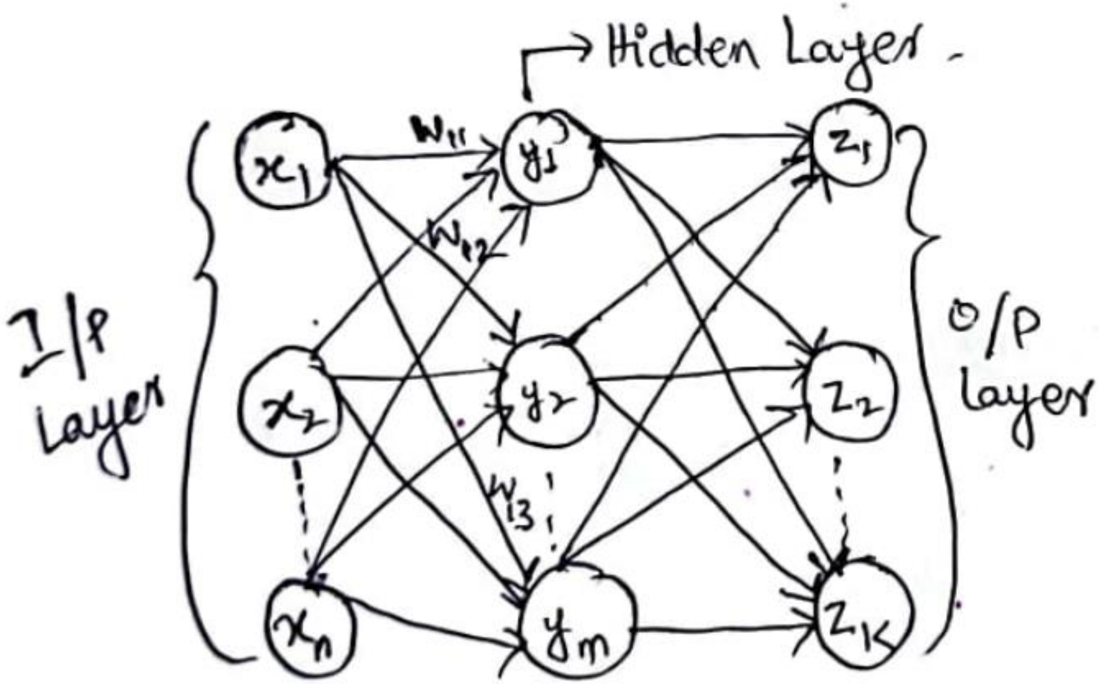
① SINGLE LAYER FEED FORWARD N/W



↳ It has only two layers i.e., I/P & O/P layer..

4. Types of Artificial Neural Network (2/4)

(2) MULTILAYER FEED FORWARD N/W

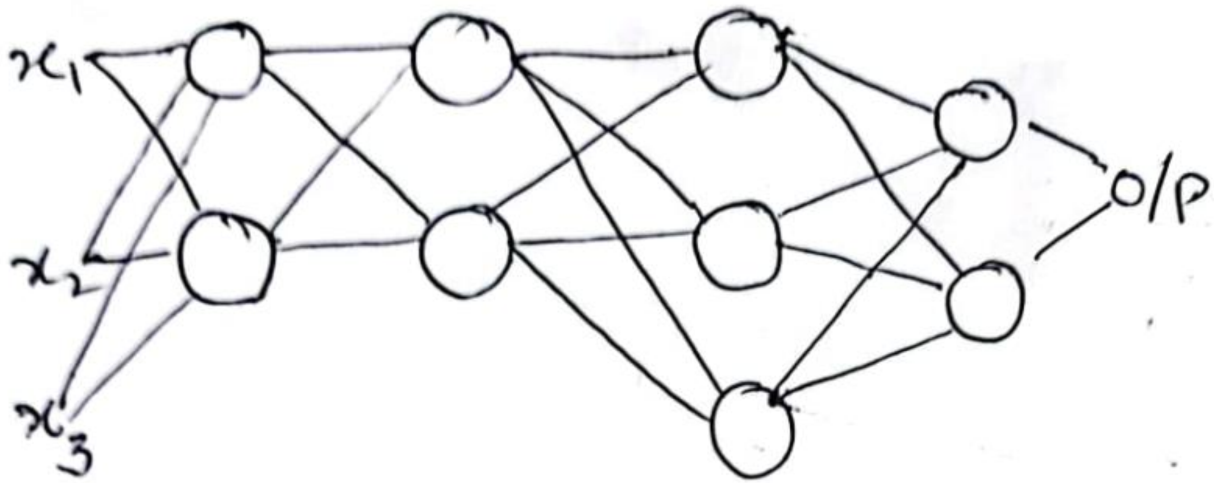


↳ It has three layers i.e., I/P, hidden & O/P layer.

↳ Inclusion of hidden layer makes system computationally more stronger.

4. Types of Artificial Neural Network (3/4)

3 MULTILAYER PERCEPTRON



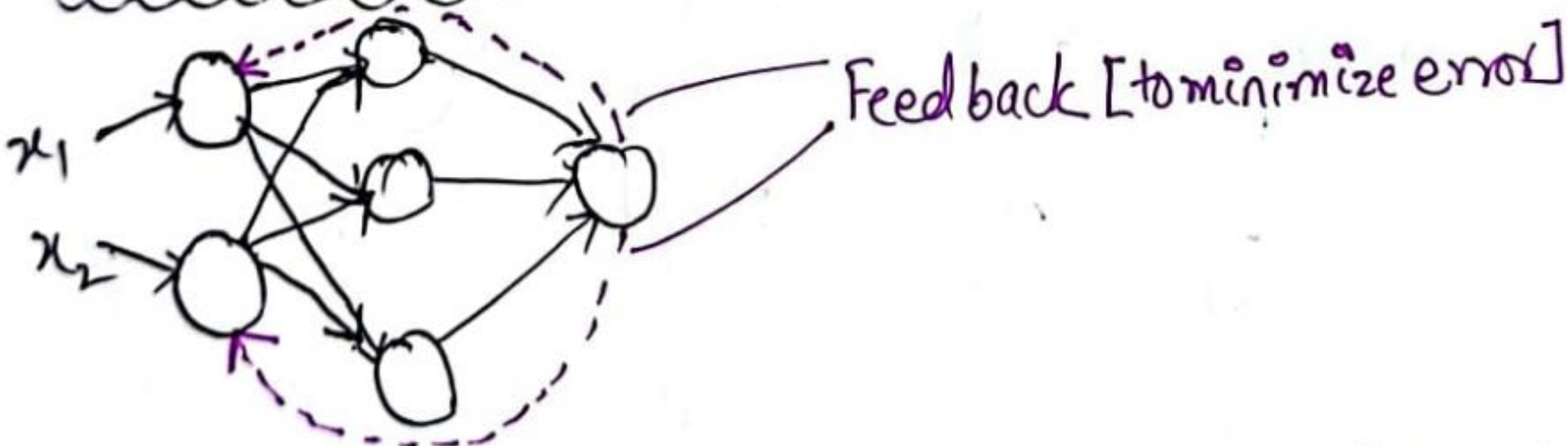
↳ 3 or more layers are used to classify non-linearly separable data.

↳ When two classes that does not seperated by decision line are known as Non-linear separable Problem.

↳ Full connected & uses non-linear Activation Function.

4. Types of Artificial Neural Network (4/4)

④ FEEDBACK ANN

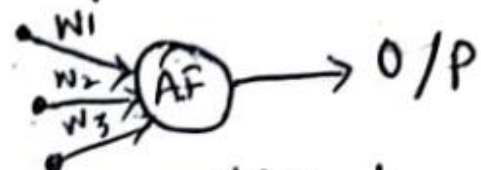


↳ Feedback is provided to input layer to adjust the parameters.

5. Activation Functions (1/3)

→ Activation function is an internal state of neuron.

↳ used to convert the I/P signal on node of ANN to an O/P signal.



↳ they introduce non-linear properties to N/W

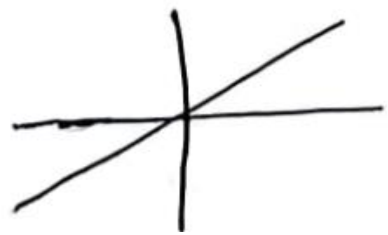
→ **Without Activation Function**: Output signal would be simple Linear Function. Easy to solve but limited in processing complex learning [N/W would fail]. Simple linear function is of polynomial degree one. In case of images, video, audio polynomial degree one function would fail.

↳ Weighted sum of input becomes input signal to A.F to give one output signal.

5. Activation Functions (2/3)

Types of Activation Functions:

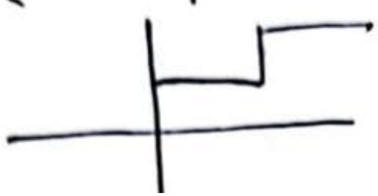
(i) **Identity Function:** $f(x) = x$ for all x. It is a linear function and output is same as input.



(ii) **Binary Step Function:**

↳ It is used in single Layer network to convert net input to output. Output can be 0 or 1.

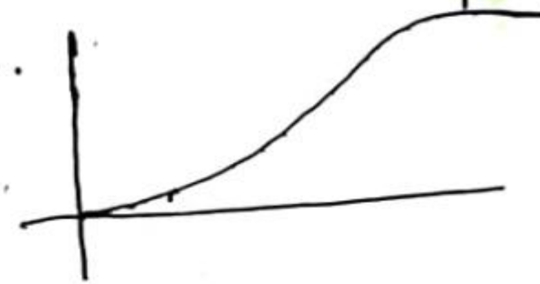
$$f(x) = \begin{cases} 1 & \text{if } x >= t \\ 0 & \text{if } x < t \end{cases}$$; where t = Threshold value



5. Activation Functions (3/3)

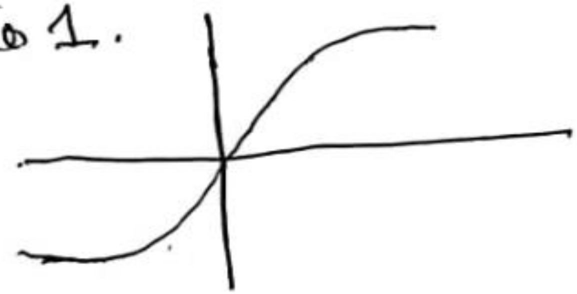
iii) **Sigmoidal Function:** It is used in back propagation networks. Range is 0 to 1.

$$f(x) = \frac{1}{1+\exp^{(-x)}} = \frac{1}{1+e^{-x}}$$



iv) **Hyperbolic Tanget Function (Tanh):** optimization is easy. Range is -1 To 1.

$$f(x) = \frac{1-e^{-2x}}{1+e^{-2x}}$$



(v) **Ramp Function:**

$$R(x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

References: Textbooks

1. *"Artificial Intelligence", by Elaine Rich and Kevin Knight, (2006), McGraw Hill companies Inc., Chapter 1-22 page 1-613.*
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3. *"Computational Intelligence: A logical Approach", by David Poole, Alan Mackworth, and Randy Goebel, (1998), Oxford University Press, Chapter 1-12, page 1-608.*
4. *"Artificial Intelligence: Structures and Strategies for Complex Problem Solving", by George F.Lugar, (2002), Addison-Wesley, Chapter 1-16, page 1-743.*
5. *"AI: A new Synthesis", by Nils J.Nilsson, (1998), Morgan Kaufmann Inc., Chapter 1-25, Page 1-493.*
6. *"Artificial Intelligence: Theory and Practice" by Thomas Dean, (1994), Addison-Wesley, Chapter 1-10, Page 1-650.*
7. *Related documents from open source, mainly internet.*